Haonan LIU

Ph.D., Assistant Professor

Department of Chemical Science and Engineering, Institute of Science Tokyo
Room 383, Main Bldg. (H-133), 2-12-1 Ookayama, Meguro, Tokyo 152-8552, Japan
liu@mct.isct.ac.jp ORCID: 0000-0001-8989-7173
h-ryu.com

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Profile

Haonan Liu is an Assistant Professor in the Department of Chemical Science and Engineering at the Institute of Science Tokyo. His research focuses on polymer analysis, particularly high-frequency dielectric properties, nanomechanics, high-pressure optical properties, and spectroscopic characterization of molecular dynamics.

Research Interests

Polymer dielectric materials for high-frequency applications (GHz–THz); low-loss polyimides; molecular dynamics–dielectric correlations; terahertz spectroscopy; atomic force microscopy and polymer nanomechanics.

Appointments

Assistant Professor, Nakajima Laboratory, Institute of Science Tokyo	Apr. 2026–Present
Assistant Professor, Ando Laboratory, Institute of Science Tokyo	Oct. 2024–Mar. 2026
Assistant Professor, Ando Laboratory, Tokyo Institute of Technology	Apr. 2023–Sept. 2024
Postdoctoral Researcher, Nakajima Laboratory, Tokyo Institute of Technology	Apr. 2022–Mar. 2023

Education

Dr. Eng., Chemical Science and Engineering, Tokyo Institute of Technology — Mar. 2022
Thesis: Morphological and nanomechanical characterization of block copolymer-based thermoplastic elastomers

M. Eng., Chemical Science and Engineering, Tokyo Institute of Technology — Mar. 2019
Thesis: Synthesis, structure, and physical properties of tripodal triptycene-appended vinylic polymers and copolymers

B. Eng., Chemical Engineering and Technology, Zhejiang University — June 2016
Thesis: Thermoresponsive gels with crystallizable side chains

Academic Service

Research Group on Computational Polymer Science, SPSJ, Japan, Organizing Committee	2026–Present
Japan Polyimide & Aromatic Polymers Conference, Organizing Committee	2023–Present

Selected Publications

- Sawada, R.[†]; **Liu, H.**[†]; Yanagimoto, S.; Yanagimoto, Y.; Ando, S. *Extremely Low Humidity and Frequency Dependence of Dielectric Properties of Highly Fluorinated Polyimides in the 10–330 GHz Range. ACS Applied Polymer Materials*, 7, 14371–14381 (2025).
- **Liu, H.**; Isoda, R.; Doi, M.; Muto, K.; Ando, S. *Pressure-Induced Enhancement of Room-Temperature Phosphorescence in Heavy Halogen-Containing Imide and Polyimide. The Journal of Physical Chemistry B*, 128, 7690–7701 (2024).
- **Liu, H.**; Sawada, R.; Yanagimoto, S.; Yanagimoto, Y.; Ando, S. *Frequency-Dependent Dielectric Properties of Aromatic Polyimides in the 25–330 GHz Range. Applied Physics Letters*, 124, 232903 (2024).
- **Liu, H.**; Liang, X.; Nakajima, K. *Nanoscale Strain–Stress Mapping for a Thermoplastic Elastomer Revealed Using a Combination of In Situ Atomic Force Microscopy Nanomechanics and Delaunay Triangulation. Journal of Polymer Science*, 60, 3134–3140 (2022).
- **Liu, H.**; Liang, X.; Nakajima, K. *Direct Visualization of a Strain-Induced Dynamic Stress Network in a SEBS Thermoplastic Elastomer with In Situ AFM Nanomechanics. Japanese Journal of Applied Physics*, 59, SN1013 (2020).